E-commerce

Customer Behavior Analysis

Exploratory analysis of user behaviour, product trends & price insights

Dataset: main.xlsx | Date: 9-04-2026| 845K Records

Dataset Overview

845041	9	view, cart, purchase	Oct 2019 – Feb 2020
Total Records	Columns	Event Types	Date Range

Dataset Columns

event_time	price
event_type	user_id
product_id	user_session
category_id	user_session
category_code	user_session
brand	user_session

Data Cleaning & Export

01

Data Loading

Loaded main.xlsx from Google Drive using pandas read_excel()

02

Exploration

Inspected 845k rows × 9 columns; reviewed dtypes and null values

03

Preprocessing

Separated event_time into date and time columns for granular analysis and deleted irrelevent columns

04

Visualisation

Plotted event distributions, category frequencies, price ranges & brands

05

Export

Saved cleaned data to cleaned_data.xlsx and downloaded via Google Colab

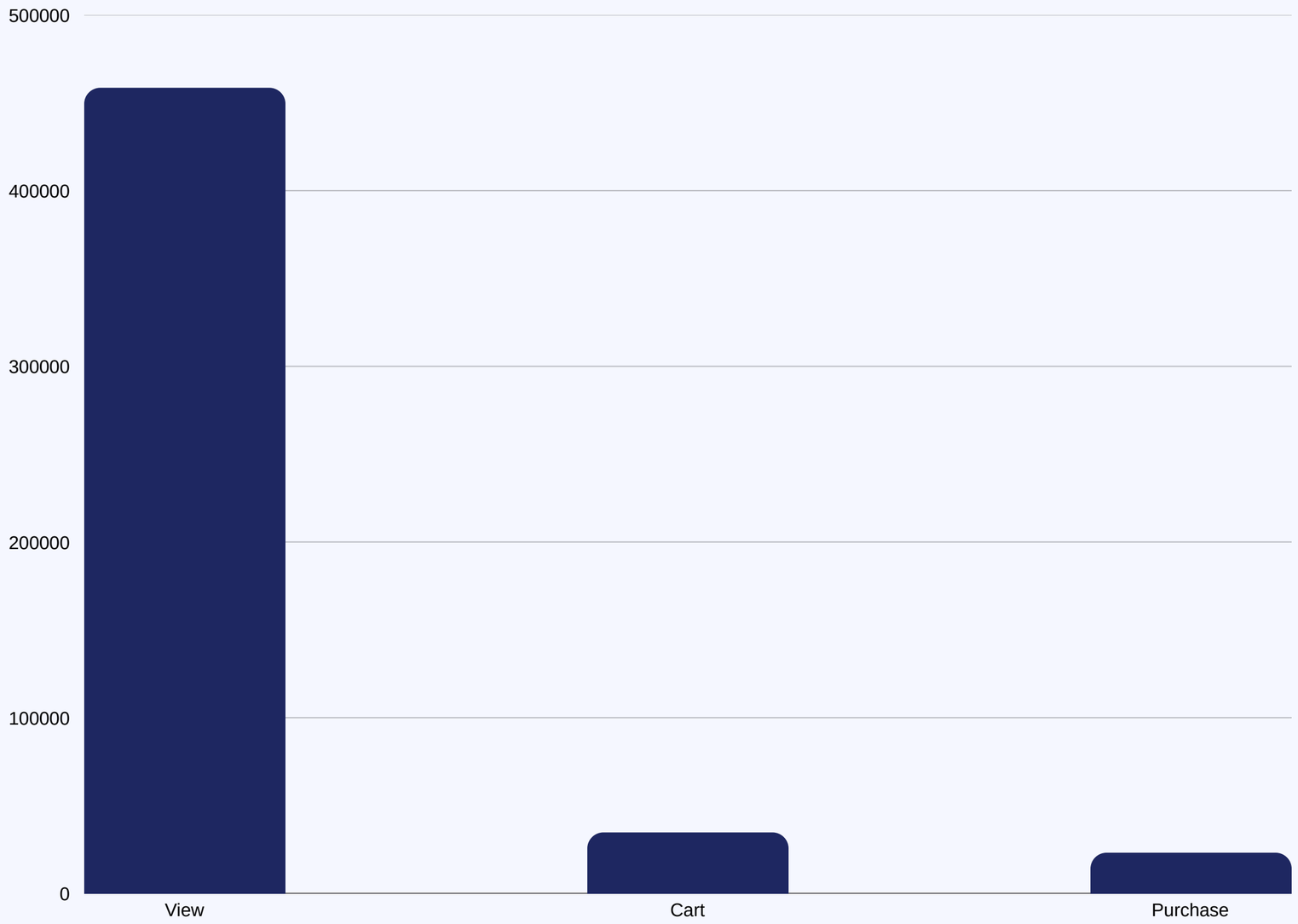
Updated Dataset Overview

516350	11	view, cart, purchase	Oct 2019 – Feb 2020
Total Records	Columns	Event Types	Date Range

Dataset Columns

- event_time
- event_type
- product_id
- category_id
- category_code
- brand
- price
- date
- day
- weekday
- month

Event Type Distribution



Key Observations

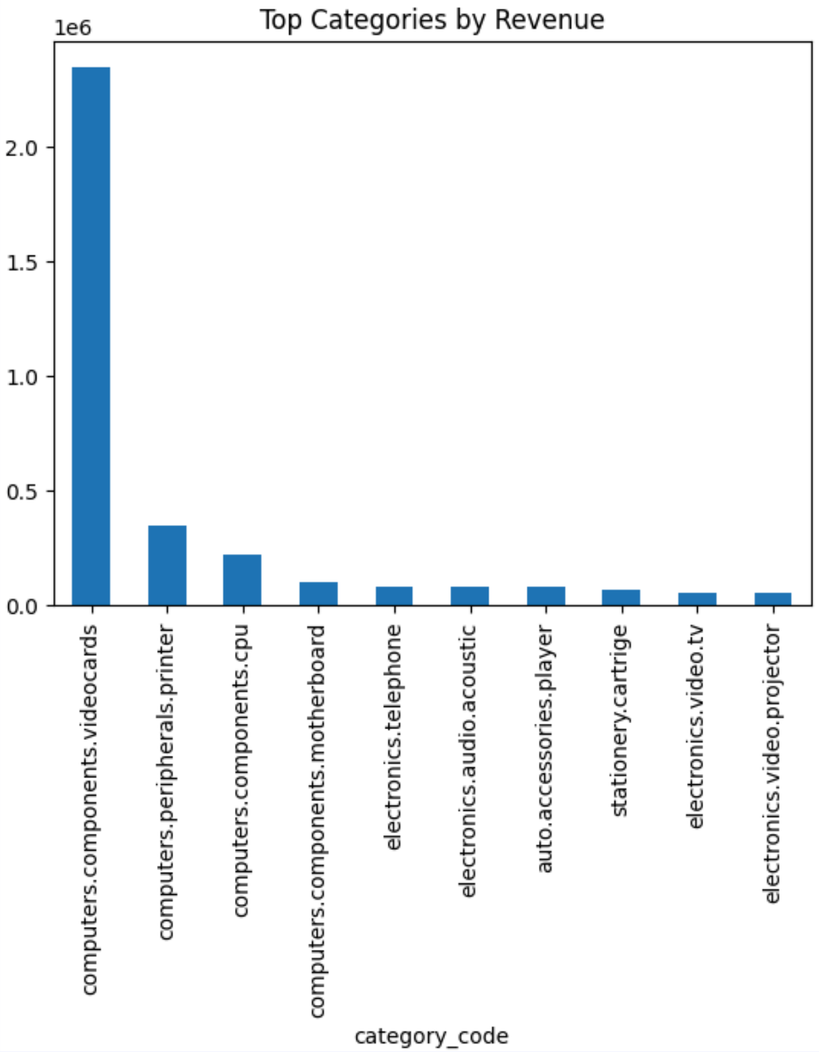
View events dominate (~88.8%) of all interactions

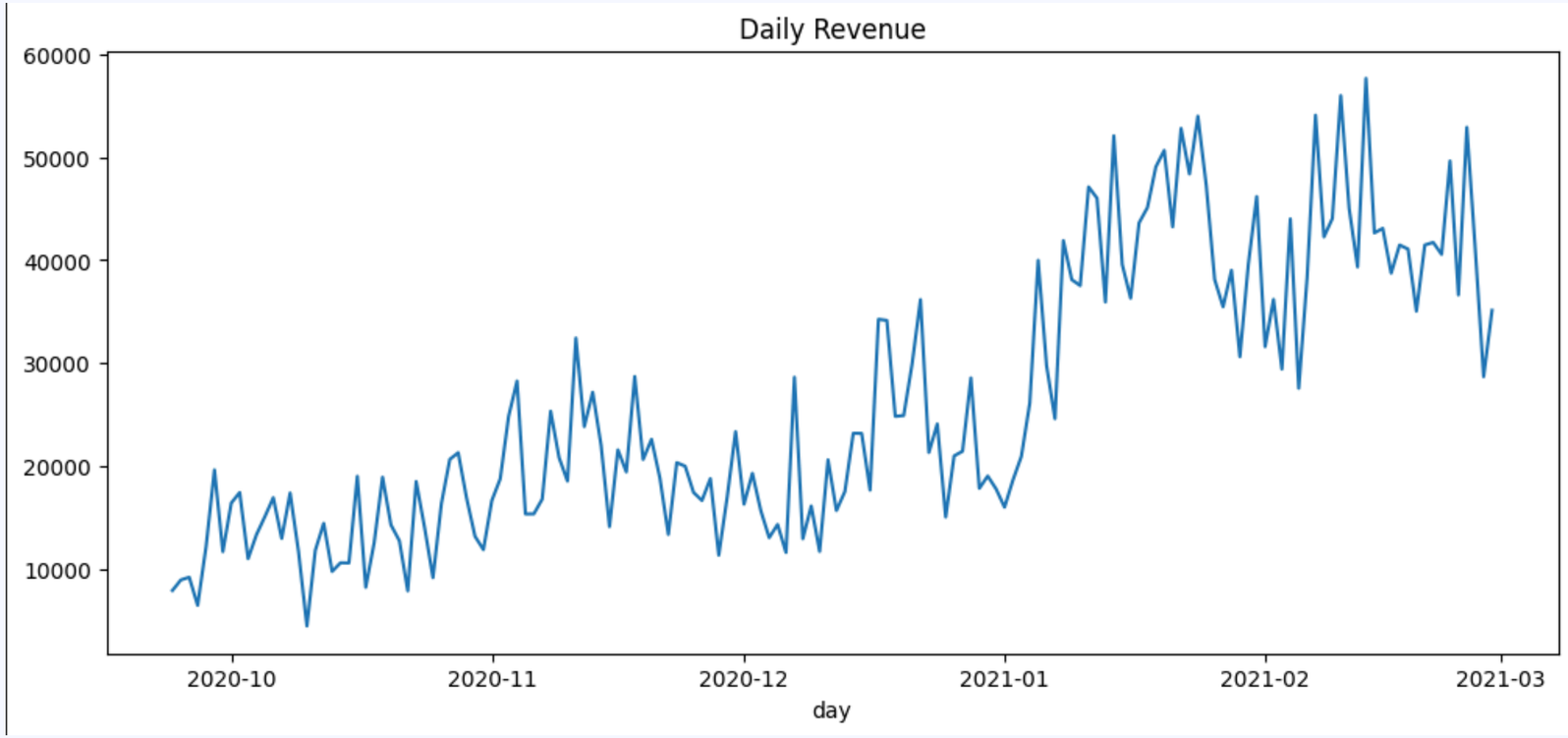
Cart adds represent (~6.7)% of events

Purchase conversions at (~4.4)%

Mostly browse-heavy session data

Top Product Categories And Daily Revenue

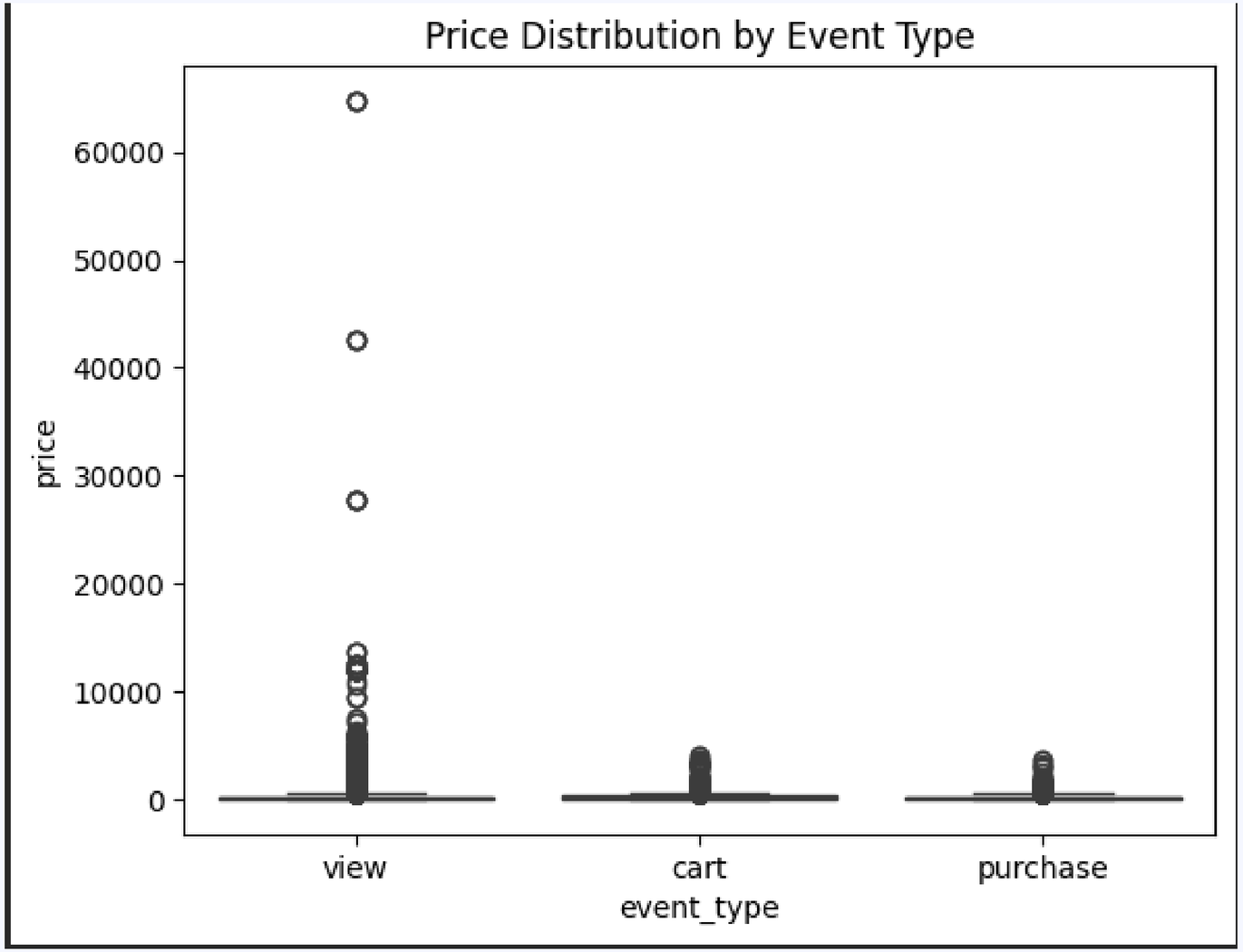




Total Revenue-4138323(approx)

Electronics & computers components lead product views in the dataset.

Price Distribution Insights



$0.90

Min Price

$3717.65

Max Price

~$179.10

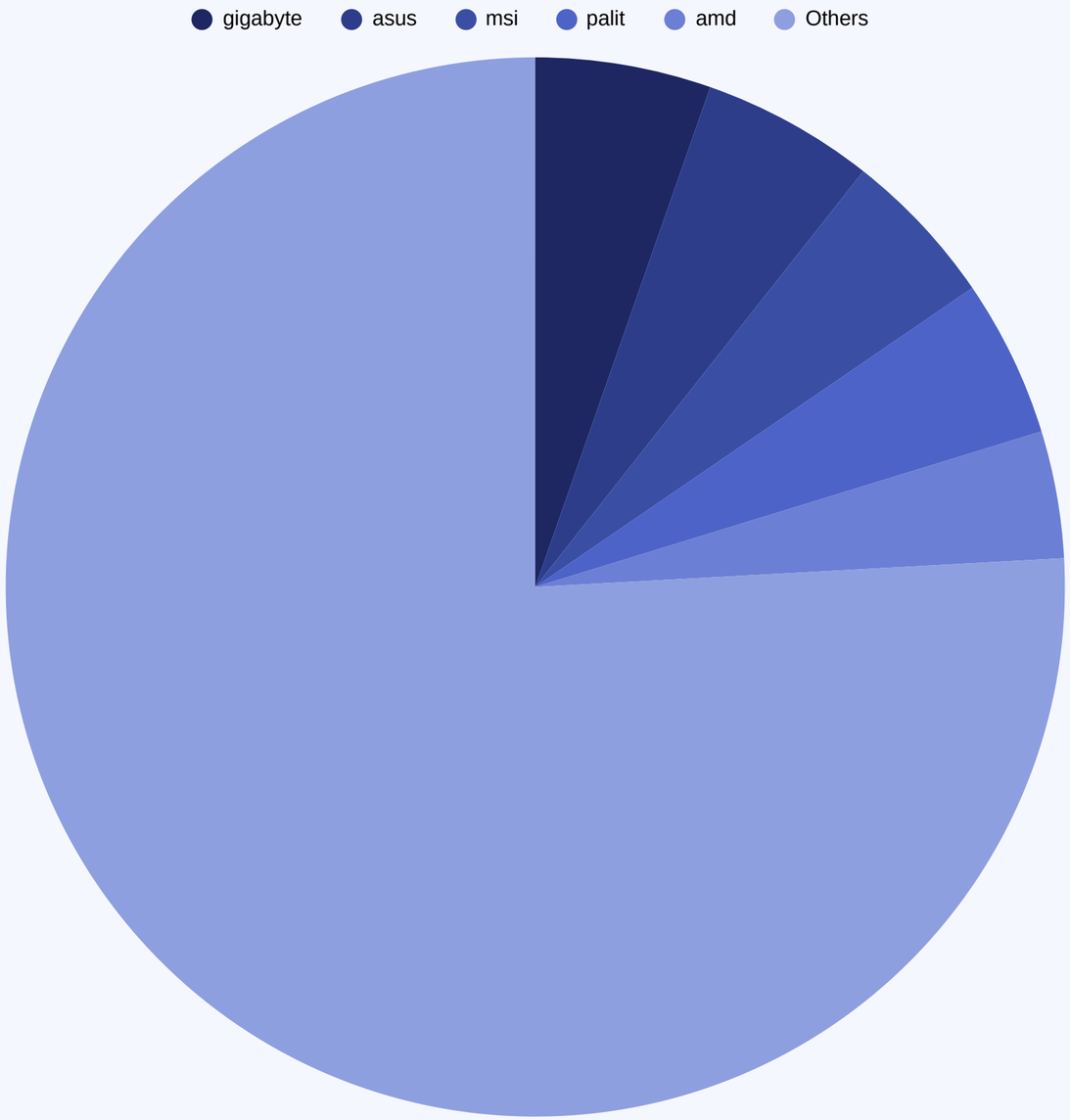
Avg Price

~$117.75

Median Price

Most common Pricr Range - $2.817-$372.575

Top Brands by Product Views



Brand Highlights

gigabyte

asus

msi

palit

amd

Key Takeaways

- ✓ Most user activity consists of product views
- ✓ **Videocards** are the most viewed product categories
- ✓ Majority of products are priced in the $2.817-$372.575 range
- ✓ Several prominent brands (gigabyte, asus) dominate category views
- ✓ Data was cleaned and exported to cleaned_data.xlsx for further analysis